

MEMORANDUM

Grid carbon intensity and data center siting

Subject: Where to site a 10 MW data center, and whether smart scheduling can substitute for a good location

Data: EIA-930 hourly generation by fuel, four US grid regions, calendar year 2025

Date: September 2026

The short version

A company siting a new 10 MW data center has two levers for cutting the carbon footprint of the electricity it buys: where it builds, and when it runs its flexible work. This analysis prices both levers using a full year of real hourly grid data.

The location lever is worth about 16,000 tonnes of CO₂ per year. The scheduling lever is worth between 450 and 1,800. Siting is not the first-order decision, it is very nearly the only decision.

Region	Mean CI	Flat 24/7	Load-shifted	Reduction
	(g/kWh)	(tCO ₂ e/yr)	(tCO ₂ e/yr)	
CAISO	231	20,223	18,958	-6.3%
ERCOT	316	27,595	25,842	-6.4%
PJM	352	30,847	30,396	-1.5%
MISO	416	36,481	35,617	-2.4%

Full-year 2025 results. Load-shifting assumes 40% of load is flexible within each day, capped at 1.5x nameplate.

1. Where you build sets the floor

Every hour of 2025 was converted into a single number, grams of CO₂ equivalent per kilowatt-hour, by weighting each fuel source in the mix by its IPCC AR5 lifecycle emissions factor. Averaged over the year, the four regions separate cleanly.

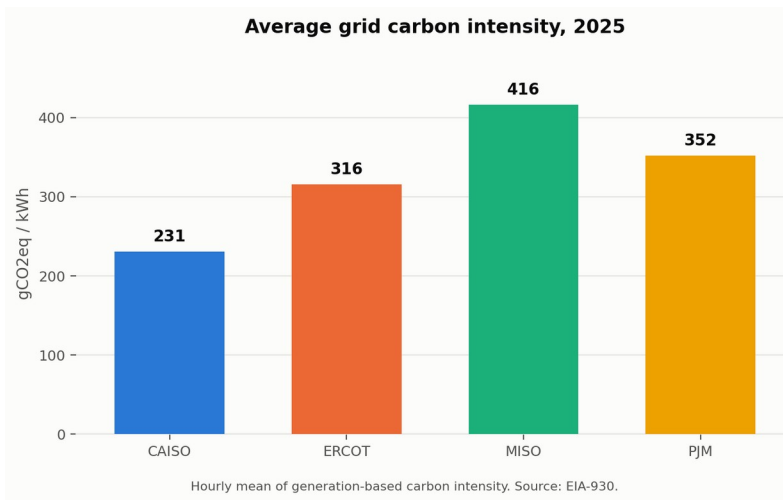


Figure 1. Average carbon intensity by region, 2025.

CAISO runs at 231 g/kWh, MISO at 416. Run the same 10 MW around the clock in each and MISO emits **1.80 times** what CAISO does, a gap of 16,258 tonnes a year for identical computation. PJM is worth noting on its own: it is the largest data center market in the world, and it ranks third of four at 352 g/kWh.

2. When you run matters, but only in some places

Carbon intensity is not a constant. In California, solar pushes it down through the middle of the day and the evening ramp onto gas pushes it back up. A sample summer week shows the shape.

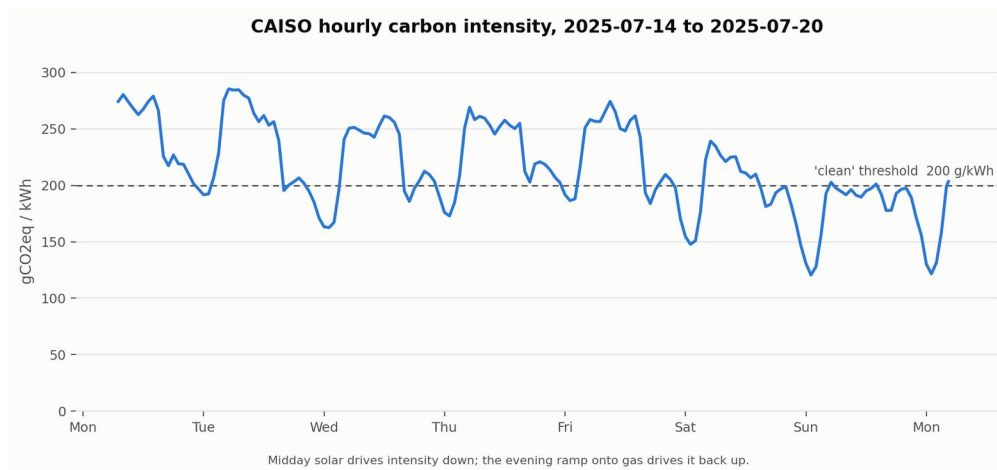


Figure 2. CAISO hourly carbon intensity, July 14 to 20, 2025.

36.7% of CAISO hours in 2025 fell below 200 g/kWh. In PJM and MISO, not a single hour did. The daily swing tells the same story: ERCOT moves 120 g/kWh between its cleanest and dirtiest average hour, CAISO 75, MISO 51, and PJM only 31. Scheduling is a real lever in Texas and California and close to a dead letter in PJM.

One result ran against expectation. CAISO's cleanest average hour is **5pm, not noon**. Gas output stays high through midday even as solar peaks, then falls faster than solar does in the late afternoon. Averaged across a full year, including low-solar winter, the late-afternoon trough wins.

3. Good scheduling does not rescue a bad location

This is the question worth asking, because it is the one a siting team actually argues about. If a company has a business reason to build in Texas or Virginia, can carbon-aware scheduling close the gap to California?

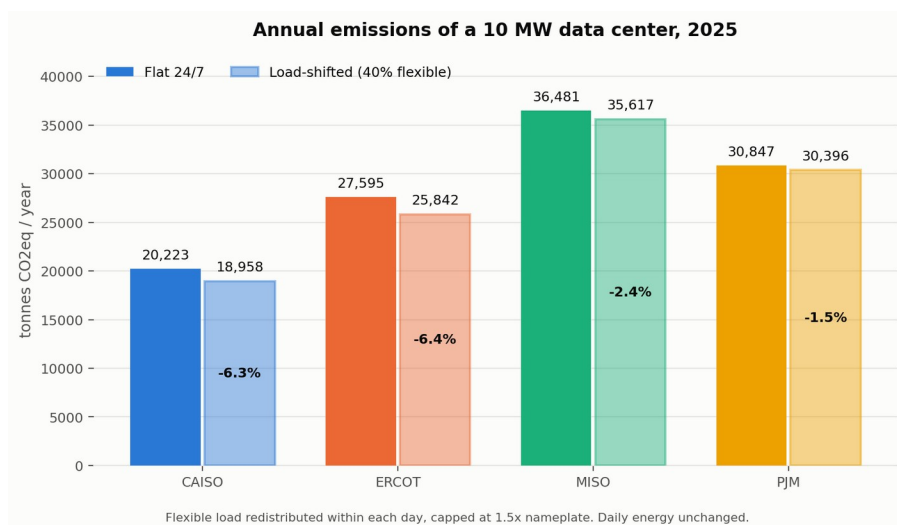


Figure 3. Flat 24/7 versus load-shifted annual emissions, 10 MW data center.

It cannot, and it is not close. ERCOT with aggressive load-shifting still emits 25,842 tCO₂e, which is **5,619 tonnes more** than CAISO does running flat and doing nothing at all. Even an unattainable upper bound, perfect scheduling across the whole year rather than within each day, leaves ERCOT at 23,568 and still behind. Scheduling is worth 1.5% to 6.4%. Siting is worth 80%.

4. The ranking holds across seasons

A ranking built on one year is only useful if it is stable. Comparing a January week against a July week, the order does not change.

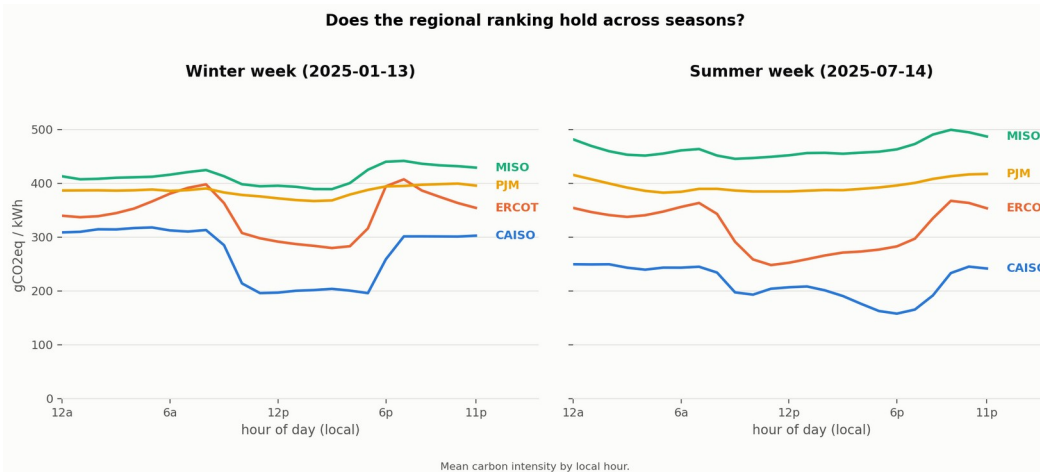


Figure 4. Mean carbon intensity by local hour, winter week versus summer week.

CAISO stays below ERCOT, which stays below PJM, which stays below MISO, in both seasons. ERCOT's winter wind narrows the midday gap to CAISO but never closes it. The lines also make the timing finding visible: CAISO and ERCOT have pronounced midday troughs, PJM is nearly flat all day.

Recommendation

- **Site in CAISO.** It is the cleanest of the four on average and it has the deepest supply of genuinely clean hours, 36.7% of the year below 200 g/kWh.
- **Treat scheduling as a second-order optimization.** Worth building in CAISO and ERCOT, where the daily swing is large. Not worth engineering effort in PJM, where the grid is nearly flat around the clock.
- **Do not use load-shifting to justify a dirtier region.** The numbers do not support it, and the average-emissions method used here is generous to load-shifting to begin with.

Method and limitations

Hourly generation by fuel came from EIA-930 for CAISO, ERCOT, PJM, and MISO across all of 2025. Each hour's fuel mix was weighted by IPCC AR5 WG3 lifecycle emissions factors to produce one carbon intensity figure per region-hour, then multiplied by a flat 10 MW load. The load-shifted case redistributes 40% of each day's energy into that day's cleanest hours, capped at 1.5x nameplate, holding daily energy constant.

One methodological note is worth flagging because it changed a headline number. CAISO reports grid batteries under the generic OTH fuel code and ERCOT under UES. A naive filter treats stored solar as fossil generation. Classifying a series as storage when its charging energy exceeds 1% of its own output caught both, and correcting it moved CAISO from 234.0 to 230.9 g/kWh.

Three limits belong on any read of these numbers. The analysis is generation-based, so imports are ignored, and CAISO is a large net importer, meaning its true consumption intensity is higher than 231 and its lead over ERCOT is overstated. It uses average rather than marginal emissions, which overstates the load-shifting benefit, a benefit that is already small. And it is one year and one weather draw.